

# Bharat Agrawal

✉ [bharat.agrawal1322@gmail.com](mailto:bharat.agrawal1322@gmail.com) · 📞 +91 9368914189

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## EDUCATION

### Indian Institute of Technology (IIT), Mandi

B.Tech. in Computer Science and Engineering (CGPA: 9.04 / 10.00)

Relevant Coursework: DSA, Database Systems, Machine Learning, Deep Learning, Computer Organization

Aug 2023 – Present

Mandi, India

### Technical University of Munich (TUM)

Exchange Semester in Informatics (Grade: 1.7/4.0, Converted GPA: 9.44/10.00)

Relevant Coursework: Graph Neural Networks, Advanced C++, Natural Language Processing

Oct 2025 – Mar 2026

Munich, Germany

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, Bash, JavaScript, SQL, HTML, CSS

**Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Flask, Django, Streamlit, Pandas, Numpy, React.js, Node.js, FastAPI, Ngrok, Selenium, BeautifulSoup, Requests

**Tools & Systems:** AWS, VSCode, Linux, Docker, Git, GitHub, Perfetto

## TECHNICAL EXPERIENCE

### Google, Software Engineering Intern

May 2026 – Aug 2026

Silicon Product Software

Bengaluru, India

- Streamlined performance analysis for Linux and Android by architecting a unified profiling pipeline that synchronizes 490+ GPU hardware counters and CPU traces for GUI visualization, reducing bottleneck identification time by 40%.
- Identified several GPU performance bottlenecks by developing custom scripts to merge kernel-space events with a newly implemented user-space API logging mechanism.
- Optimized GPU Driver by tweaking heap memory allocation, reducing OOM stall cycles by 60%.
- Enhanced the GPU pipeline by replacing synchronous cache flushes with deferred background cache invalidations.
- Patched the ANGLE trace test to support headless **Vulkan** trace replays on Debian based platforms.

## RESEARCH AND PROJECTS

### Tabularis - Table Understanding and Query System

[GitHub](#)

- Developed a pipeline consisting of (Preprocessor - Pix2Struct (240M) - Qwen 3.0) to extract tables from PDFs and Images and converting them to HTML and CSV Format.
- Processed 1000+ documents, achieving **95%+ accuracy** in table recognition and data extraction.
- Integrated **Qwen 3.0** with memory, enabling table queries across 50+ diverse types achieving 90% resolution rate.

### SemEval-2026 Task-12: Abductive Event Reasoning for Large Language Models

[Paper](#)

- Built a **dual filtering pipeline** (BM25 + FAISS re-ranking) to extract causal evidence from large document collections.
- Fine-tuned Llama 3.2 3B and 8B using 4-bit quantization (Unsloth, ~0.75% params trained).
- Llama 3.2 8B hit **accuracy 0.80, precision 0.90, F1 0.88**, well above all few-shot prompting baselines.

### Deep Learning Models in Local Feature Matching

[Paper](#)

- Conducted a survey of **10 papers** and 30+ supporting works spanning **GNN sparse matchers** and **transformer dense approaches**.
- Proposed a 3-dimensional taxonomy based on feature modality, architectural strategy, and learning paradigm.
- Performed an evaluation on 2 benchmarks - **MegaDepth** and **ScanNet** using **pose error and AUC metrics**.

### Automatic Modulation Recognition

[GitHub](#)

- Implemented the **VTCNN2** model using the **Radio ML 2016.10a** Dataset, achieving an **overall accuracy of 72%** in classifying **8 Digital** and **3 Analog** signal types of SNR [0,18].
- Proposed a DNN consisting of CNN, LSTM, GRU layers, enhancing classification accuracy to **92%**.
- Reduced model complexity, decreased inference latency by **40%** with an average response time of **50 ms**.

## ACHIEVEMENTS

- Secured **2nd** place in HCLTech Hackathon organized under CS671 course.
- Ranked **10th nationwide** among 50+ teams in CSAW CTF.
- Attained **4th** rank in Scythe CTF and **6th** rank in Chronos CTF.
- Solved **1100+** Coding problems across platforms like Codeforces, LeetCode, CodeChef, GFG.

## POSITIONS OF RESPONSIBILITY

- Mentor, **Kamand Prompt** - Programming Club of IIT Mandi
- Mentor, **Association for Computing Machinery (ACM)**